TASK 7

Create User Defined Functions/Eval functions in Pig to handle unwanted data during data processing.

w.txt:

```
01/01/2010,12.4,18.7
05/02/2010,10.5,25.3
10/06/2011,20.3,32.1
15/08/2011,22.1,35.6
20/04/2012,15.7,28.3
25/12/2012,-5.0,30.8      <-- INVALID (Negative min temp)
30/07/2013,26.8,40.5
05/11/2013,10.2,25.9
12/09/2014,24.5,38.7
18/10/2014,50.0,41.2      <-- INVALID (Max temp < Min temp)
,,                        <-- INVALID (Missing values)
23/05/2015,NaN,32.0       <-- INVALID (NaN min temp)
30/07/2015,22.0,          <-- INVALID (Missing max temp)
01/01/2016,15.0,NaN       <-- INVALID (NaN max temp)
15/06/2016,19.4,39.7
22/11/2016,-3.5,28.2      <-- INVALID (Negative min temp)
10/08/2017,26.3,42.5
31/12/2017,29.1,45.0
04/07/2018,24.0,38.4
20/10/2018,21.8,37.2
```

cw.py

```
@outputSchema("cleaned_record: tuple(date:chararray, min_temp:float, max_temp:float)")
def clean_data(date, min_temp, max_temp):
    try:
        if not date or not min_temp or not max_temp:
            return None

        min_temp = float(min_temp)
        max_temp = float(max_temp)

        if min_temp != min_temp or max_temp != max_temp:
            return None

        if min_temp < 0 or max_temp < 0:
            return None

        if max_temp < min_temp:
            return None
```

```
        return (date, min_temp, max_temp)

    except (ValueError, TypeError):
        return None
```

cw.pig

```
REGISTER 'cw.py' USING jython AS weather_udf;

weather_data = LOAD 't7/w.txt' USING PigStorage(',')
    AS (date:CHARARRAY, min_temp:CHARARRAY, max_temp:CHARARRAY);

filtered_data = FOREACH weather_data GENERATE weather_udf.clean_data(date, min_temp, max_temp) AS cleaned_record;

valid_data = FILTER filtered_data BY cleaned_record IS NOT NULL;

cleaned_weather = FOREACH valid_data GENERATE cleaned_record.date, cleaned_record.min_temp, cleaned_record.max_temp;

STORE cleaned_weather INTO 't7/c' USING PigStorage(',');
```

Run the script: pig -x mapreduce cw.pig

Output:

```
[cloudera@quickstart Desktop]$ pig -x mapreduce cw.pig
log4j:WARN No appenders could be found for logger (org.apache.hadoop.util.Shell)
.
log4j:WARN Please initialize the log4j system properly.
log4j:WARN See http://logging.apache.org/log4j/1.2/faq.html#noconfig for more in
fo.
2025-03-20 02:32:18,183 [main] INFO  org.apache.pig.Main - Apache Pig version 0.
12.0-cdh5.13.0 (rexported) compiled Oct 04 2017, 11:09:03
2025-03-20 02:32:18,184 [main] INFO  org.apache.pig.Main - Logging error message
s to: /home/cloudera/Desktop/pig_1742463138165.log
2025-03-20 02:32:19,106 [main] INFO  org.apache.pig.impl.util.Utils - Default bo
otup file /home/cloudera/.pigbootup not found
2025-03-20 02:32:19,232 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.addr
ess
2025-03-20 02:32:19,233 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - fs.default.name is deprecated. Instead, use fs.defaultFS
2025-03-20 02:32:19,233 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.HExecutionEngine - Connecting to hadoop file system at: hdfs://quickstart.clo
udera:8020
2025-03-20 02:32:19,817 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.HExecutionEngine - Connecting to map-reduce job tracker at: localhost:8021
2025-03-20 02:32:19,852 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
```

```
Output(s):
Successfully stored 13 records (273 bytes) in: "hdfs://quickstart.cloudera:8020/
user/cloudera/t7/c2"

Counters:
Total records written : 13
Total bytes written : 273
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1742461679584_0008


2025-03-20 02:32:47,260 [main] WARN  org.apache.pig.backend.hadoop.executionengi
ne.mapReduceLayer.MapReduceLauncher - Encountered Warning ACCESSING_NON_EXISTENT
_FIELD 42 time(s).
2025-03-20 02:32:47,260 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.mapReduceLayer.MapReduceLauncher - Success!
[cloudera@quickstart Desktop]$ hdfs dfs -ls t7/c2
Found 2 items
-rw-r--r--   1 cloudera cloudera          0 2025-03-20 02:32 t7/c2/_SUCCESS
-rw-r--r--   1 cloudera cloudera        273 2025-03-20 02:32 t7/c2/part-m-00000
```

```
[cloudera@quickstart Desktop]$ hdfs dfs -cat t7/c2/*
01/01/2010,12.4,18.7
05/02/2010,10.5,25.3
10/06/2011,20.3,32.1
15/08/2011,22.1,35.6
20/04/2012,15.7,28.3
30/07/2013,26.8,40.5
05/11/2013,10.2,25.9
12/09/2014,24.5,38.7
15/06/2016,19.4,39.7
10/08/2017,26.3,42.5
31/12/2017,29.1,45.0
04/07/2018,24.0,38.4
20/10/2018,21.8,37.2
[cloudera@quickstart Desktop]$
```